Algebraic derivation of semi-small decomposition of the Chow ring of a matroid

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1 Introduction

Introduce the context.

2 Algebraic Framework

2.1 Incidence algebra and factorizations

Let P be a graded bounded poset with rank function $\mathrm{rk} : P \to \mathbb{Z}_{\geqslant 0}$. The *incidence algebra* $I(P, R)$ of P over the commutative ring R consists of all functions $h : P \times P \to R$ such that

$$h(s,t) = 0 \text{ unless } s \leqslant t.$$

With appropriate indexing, we view these functions as $|P| \times |P|$ upper triangular matrices with entries $h_{s,t} = h(s,t)$ in R. The product on $I(P,R)$ is given by matrix multiplication. In our case, we will focus on $\mathcal{I} := I(P, \mathbb{Z}[x])$ which admits an involution $(\cdot)^{\mathrm{rev}} : \mathcal{I} \to \mathcal{I}$ given by

$$h^{\mathrm{rev}}_{st}(x) = x^{\mathrm{rk}(t)-\mathrm{rk}(s)} h_{st}(x^{-1}).$$

The incidence algebra works as a hub for all the polynomial invariants of the poset P. The Kazhdan–Lusztig–Stanley function f and the Chow function H are defined as unique functions in the incidence algebra satisfying certain conditions.

Definition 2.1. For every poset P, there exists a unique function $f \in \mathcal{I}$ that satisfies

- $f_{ss}(x) = 1$,
- $\deg(f_{st}(x)) < \frac{\mathrm{rk}(t)-\mathrm{rk}(s)}{2}$ for all $s < t$,
- $Z := \zeta^{\mathrm{rev}} f$ satisfies $Z^{\mathrm{rev}} = Z$.

Definition 2.2. For every poset P, there exists a unique function $H \in \mathcal{I}$ that satisfies

- $H_{ss}(x) = 1$,
- $H_{st}(x) = x^{\mathrm{rk}(t)-\mathrm{rk}(s)-1} H_{st}(x^{-1})$ for all $s < t$,
- $G := \zeta^{\mathrm{rev}} H$ satisfies $G^{\mathrm{rev}} = G$.

Let $R := \mathbb{Z}[x, x^{-1}]$ be the ring of Laurent polynomials with integer coefficients. We define an injective map $(\cdot)^{\diamond} : \mathcal{I} \to I(P, R)$ as follows.

$$h^{\diamond}_{s,t}(x) = x^{\mathrm{rk}(t)-\mathrm{rk}(s)} h_{s,t}(x^{-2}).$$

Remark 2.3. This map is an algebra homomorphism, i.e. it satisfies $(gh)^\diamond = g^\diamond h^\diamond$, and is invertible on its image. For any integer n, we denote by $\mathrm{Pal}(n)$ the additive group of polynomials $p(x) \in R$ satisfying $p(x) = x^n p(x^{-1})$. While $\mathrm{Pal}(0)$ is a subring of R, we note that $\mathrm{Pal}(n)$ for $n \neq 0$ is not, as $\mathrm{Pal}(n) \cdot \mathrm{Pal}(m) \subseteq \mathrm{Pal}(n+m)$.

Lemma 2.4. *Every Laurent polynomial* $p(x) \in R$ *can be written as* $p(x) = p_0(x) + p_2(x)$ *where* $p_0(x) \in \mathrm{Pal}(0)$ *and* $p_2(x) \in \mathrm{Pal}(2)$.

Proof. For each $k \geqslant 0$, the $(2k+1)$-dimensional vector space $R_{\leqslant k}$ of Laurent polynomials with exponents in $[-k, k]$ has complementary subspaces spanned by $\{1, x^{-1}+x, \ldots, x^{-k}+x^k\}$ and $\{x, 1+x^2, x^{-1}+x^3, \ldots, x^{-k+2}+x^k\}$, which are $\mathrm{Pal}(0)$ and $\mathrm{Pal}(2)$ respectively. □

With the diamond map and the $\mathrm{Pal}(0)$ decomposition established, we note the following properties of f and H:

- $f^\diamond_{ss} = 1 \in \mathbb{Z}$ and $f^\diamond_{st}(x) \in x\mathbb{Z}[x]$ for all $s < t$.
- $H^\diamond_{ss} = 1 \in \mathbb{Z} \subset \mathrm{Pal}(0)$ and $H^\diamond_{st}(x) \in \mathrm{Pal}(2)$ for all $s < t$.

These properties lead to the following factorizations:

Lemma 2.5. *If* $h \in \mathcal{I}$ *satisfies* $(\zeta^{\mathrm{rev}} h)^\diamond \in I(P, \mathrm{Pal}(0))$, *then* $h^\diamond = H^\diamond \cdot a^\diamond$ *where* $a^\diamond$ *is the entry-wise* $\mathrm{Pal}(0)$ *projection of* $h^\diamond$. *If* $h \in \mathcal{I}$ *further satisfies* $h^\diamond_{st}(x) \in \mathbb{Z}[x]$ *for all* $s \leqslant t$, *then* $h^\diamond = f^\diamond \cdot a^\diamond$ *where* $a^\diamond_{st} = h^\diamond_{st}(0)$ *for all* $s \leqslant t$.

2.2 The column space

Let $\mathcal{H}(P)$ be the free R-module of column vectors indexed by the elements of P. The standard basis of $\mathcal{H}(P)$ is given by the indicator vectors $\{e_t\}_{t \in P}$. Any matrix in $I(P, R)$ acts naturally on $\mathcal{H}(P)$ by left multiplication. Using the diamond map, we can push elements of $\mathcal{I}$ forward to act on this column space.

Definition 2.6. For $t \in P$, the *Kazhdan–Lusztig basis element* kl^t is the t-th column of the matrix $f^\diamond$, i.e., $kl^t := f^\diamond e_t$. Similarly, the *Chow basis element* c^t is the t-th column of the matrix $H^\diamond$, i.e., $c^t := H^\diamond e_t$.

Definition 2.7. The *perverse submodule* $\mathcal{H}_p(P)$ consists of all column vectors $v \in \mathcal{H}(P)$ satisfying two conditions:

1. $v \in (\mathbb{Z}[x])^{|P|}$ (its entries are polynomials in x).
2. $(\zeta^{\mathrm{rev}})^\diamond v \in (\mathrm{Pal}(0))^{|P|}$ (its evaluation yields palindromic polynomials of degree zero).

The relaxed submodule $\mathcal{H}'_p(P)$ consists of all vectors $v \in \mathcal{H}(P)$ satisfying only the second condition.

Because $\mathrm{Pal}(0)$ is a ring, $\mathcal{H}'_p(P)$ is a $\mathrm{Pal}(0)$-module. Since multiplying by an element of $\mathbb{Z}$ preserves both polynomiality and $\mathrm{Pal}(0)$ symmetry, $\mathcal{H}_p(P)$ is a $\mathbb{Z}$-module.
The algebraic factorizations in Theorems 2.5 translate directly into change-of-basis theorems for these column spaces. If we treat a vector $v \in \mathcal{H}'_p(P)$ as a column of a larger matrix $h^\diamond$, the incidence algebra factorizations isolate the coefficients into the required rings.

Theorem 2.8. *The Kazhdan–Lusztig elements* $\{kl^t\}_{t \in P}$ *form a* $\mathbb{Z}$*-basis for* $\mathcal{H}_p(P)$.

Theorem 2.9. *The Chow elements* $\{c^t\}_{t \in P}$ *form a* $\mathrm{Pal}(0)$*-basis for* $\mathcal{H}'_p(P)$. *Furthermore, for any* $v = \sum \alpha_s e_s \in \mathcal{H}'_p(P)$, *its expansion in the Chow basis is* $v = \sum (\alpha_s)_0 c^s$, *where* $(\alpha_s)_0$ *is the* $\mathrm{Pal}(0)$ *projection of the standard coordinate* α_s.

3 Deletion in Geometric Lattices

We now restrict our attention to the case where the poset P is the lattice of flats $\mathcal{L}(M)$ of a matroid M on a finite ground set E. For any element $e \in E$, the deletion $M \setminus e$ is a matroid on $E \setminus \{e\}$. The map $\pi : \mathcal{L}(M) \to \mathcal{L}(M \setminus e)$ defined by $\pi(F) = F \setminus \{e\}$ is surjective.

Definition 3.1. The *discrepancy* of a flat $F \in \mathcal{L}(M)$ with respect to e is the integer

$$\delta(F) := \mathrm{rk}_M(F) - \mathrm{rk}_{M \setminus e}(F \setminus \{e\}).$$

We define a linear map between the column spaces that reflects this geometric deletion.

Definition 3.2. The *deletion operator* $\Delta : \mathcal{H}(M) \to \mathcal{H}(M \setminus e)$ is the R-linear map defined on the standard basis by

$$\Delta(e_F) := x^{-\delta(F)} e_{F \setminus \{e\}}.$$

The fundamental property of Δ is that it intertwines with the evaluation map (left-multiplication by $(\zeta^{\mathrm{rev}})^\diamond$) by tracking the closure operator in M.

Lemma 3.3. *Let $H \in \mathcal{L}(M \setminus e)$ and let $\overline{H}$ be the closure of H in M. For any vector $v \in \mathcal{H}(M)$, the H-th coordinate of the evaluation of $\Delta(v)$ matches the $\overline{H}$-th coordinate of the evaluation of v:*

$$\left((\zeta^{\mathrm{rev}}_{M \setminus e})^\diamond \Delta(v) \right)_H = ((\zeta^{\mathrm{rev}}_M)^\diamond v)_{\overline{H}}$$

Theorem 3.4. *The deletion operator restricts to a $\mathrm{Pal}(0)$-module homomorphism $\Delta : \mathcal{H}'_p(P)(M) \to \mathcal{H}'_p(P)(M \setminus e)$. In particular, for any flat $F \in \mathcal{L}(M)$, $\Delta(c^F) \in \mathcal{H}'_p(P)(M \setminus e)$.*

3.1 Expansion of the Deletion Image

To compute the explicit expansion of $\Delta(c^F)$ in the Chow basis of $\mathcal{H}(M \setminus e)$, we must analyze the standard coordinates of the deleted vector. By Theorem 2.9, the coefficient of c^G in this expansion is exactly the $\mathrm{Pal}(0)$ projection of the G-th standard coordinate of $\Delta(c^F)$.
First, we define a specific subset of flats that will index our summation.

Definition 3.5. For any flat $F \in \mathcal{L}(M)$ containing e, we define the set

$$S_e(F) := \{G \in \mathcal{L}(M \setminus e) \mid G \leqslant F \setminus \{e\} \text{ and } G \cup \{e\} \in \mathcal{L}(M)\}.$$

Theorem 3.6. *For any flat $F \in \mathcal{L}(M)$, the following identities hold in $\mathcal{H}'_p(P)(M \setminus e)$:*

1. *If e is not a coloop in M^F, then*
$$\Delta(c^F) = c^{F \setminus \{e\}} + \sum_{G \in S_e(F)} x^{-1} c^F_{G \cup \{e\}} c^G \tag{1}$$

2. *If e is a coloop in M^F, then*
$$\Delta(c^F) = (x + x^{-1}) c^{F \setminus \{e\}} + \sum_{\substack{G \in S_e(F) \\ G \neq F \setminus \{e\}}} x^{-1} c^F_{G \cup \{e\}} c^G \tag{2}$$

Proof. Let $\Delta(c^F) = \sum \beta_G e_G$ be the expansion in the standard basis, and $\Delta(c^F) = \sum q_G c^G$ be the expansion in the Chow basis. By Theorem 2.9, $q_G = (\beta_G)_0$.
Applying the definition of Δ to $c^F = \sum_{L \leqslant F} c^F_L e_L$, the standard coordinate β_G is the sum over all preimages of G:

$$\beta_G = \sum_{\substack{L \leqslant F \\ L \setminus \{e\} = G}} x^{-\delta(L)} c^F_L$$

We determine q_G by analyzing the preimages $L \in \{\overline{G}, G \cup \{e\}\}$.
Case 1: $G \neq F \setminus \{e\}$**.** If $G \in S_e(F)$, both $\overline{G} = G$ and $G \cup \{e\}$ are valid flats in M. The standard coordinate is $\beta_G = c^F_G + x^{-1} c^F_{G \cup \{e\}}$. Since $c^F_G \in \mathrm{Pal}(2)$ and $x^{-1} c^F_{G \cup \{e\}} \in \mathrm{Pal}(0)$, the projection yields $q_G = x^{-1} c^F_{G \cup \{e\}}$. If $G \notin S_e(F)$, the only valid preimage is $\overline{G}$. Thus, $\beta_G = c^F_{\overline{G}} \in \mathrm{Pal}(2)$, which projects to $q_G = 0$.
Case 2: $G = F \setminus \{e\}$**.** If e is not a coloop in M^F, the only preimage is $L = F$. Thus $\beta_{F \setminus \{e\}} = x^0 c^F_F = 1 \in \mathrm{Pal}(0)$, giving $q_{F \setminus \{e\}} = 1$. If e is a coloop in M^F, the preimages are F and $F \setminus \{e\}$. Thus
$\beta_{F \setminus \{e\}} = c^F_{F \setminus \{e\}} + x^{-1} c^F_F = x + x^{-1} \in \mathrm{Pal}(0)$, giving $q_{F \setminus \{e\}} = x + x^{-1}$. □